

Question:

Let the domain be:

$$D = \{2, 3, 4, 5, 6, 7, 8\}$$

Let the following predicates be defined on D :

$P(x)$: x is prime

$Q(x)$: x is odd

Required:

- a) Write the following statement and determine its truth value, with justification

$$\forall x \in D (P(x) \rightarrow Q(x))$$

- b) Write the following statement and determine its truth value, with justification

$$\exists x \in D (\neg P(x) \rightarrow Q(x))$$

- c) Explain each of part (a) and (b) in formal English

Solution:

a) statement: $\forall x \in D (P(x) \rightarrow Q(x))$

Truth value: False

Justification: $2 \in D (P(2) \rightarrow Q(2))$
is false (2 is even and prime, its not odd)

b) statement: $\exists x \in D (\neg P(x) \rightarrow Q(x))$

Truth value: False

Justification: The only not prime numbers in the domain are:
 $\{4, 6, 8\}$ and all of them are even.

c) part(a): For all x in the domain D
if x is prime then x is odd

part(b): There is exist x in the domain D
if x is not prime then x is odd



Question:

Let the domain be:

$$D: \{1, 2, 3, 4, 5, 6, 7, 8\}$$

Let the following predicates be defined on D :

$P(x)$: x is even

$Q(x)$: $x > 5$

Required:

a) Write the following logical statement, and determine its truth value, with justification:

$$\exists x \in D \ P(x)$$

b) Write the following logical statement, and determine its truth value, with justification:

$$\exists x \in D. (P(x) \wedge Q(x))$$

c) Explain each of parts (a) and (b) in formal exam-style English

Solution:

a)

statement: $\exists x \in D P(x)$

Truth value: True

Justification: For $2 \in D P(2)$ is true since 2 is even

b) statement: $\exists x \in D (P(x) \wedge Q(x))$

Truth value: True

Justification: $6 \in D (P(6) \wedge Q(6))$ is true

$\Rightarrow 6$ is even and $6 > 5$

c) Part (a): There is exist x in the domain D such that x is even

Part (b): There is exist x in the domain D such that x is even and x is greater than 5